

Hydraulic Cabinet HSMS 850/10-6/2**Table of Contents****Designation****BA - Number**

Technical Data

BA 911262-00-4

Safety

BA 911120-00-4

Structure and Function

BA 911158-01-4

Hydraulic Cabinet HSMS 850

Technical Data

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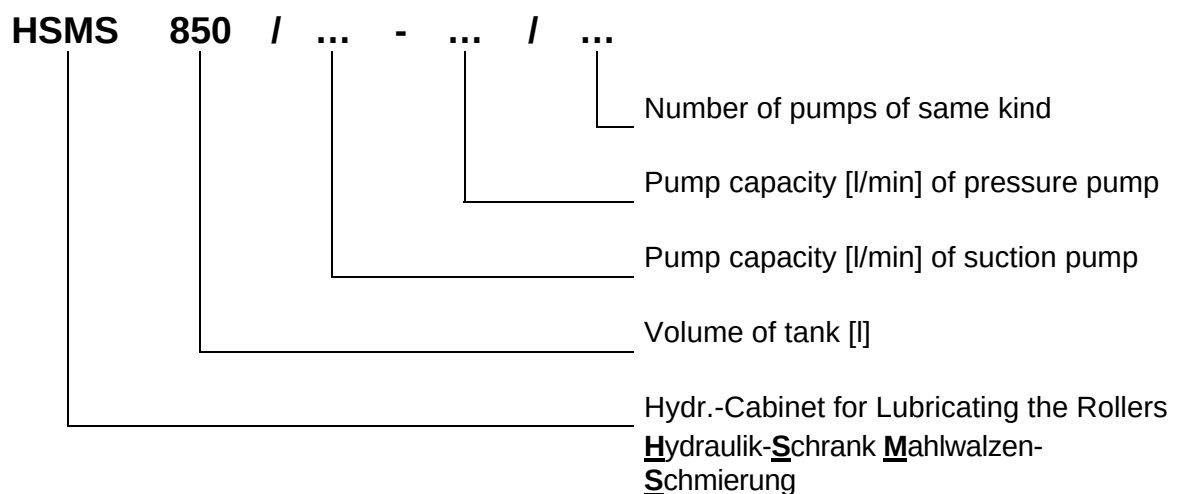
1. General

The hydraulic cabinet HSMS 850 guarantees the oil circulating for the rollers of the LOESCHE-mill.

2. Identification

Machine-Type	:	HSMS 850 ¹⁾
Machine-Partslist/-No.	:	 ¹⁾
Machine-Weight	:	 kg ¹⁾
Machine-Drawing	:	see documentation of manufacturer

3. Machine-Type Designation



¹⁾ details: see parts list of hydraulic cabinet HSMS

Hydraulic Cabinet HSMS

Safety

1. General

These operating instructions HSMS - "Safety" is a supplement to the Roller Grinding Mill - "Safety" instructions. It must be ensured that these operating instructions will be studied in detail prior to use of the hydraulic cabinet.

2. Notes on Safety at Work

- Maintenance work on the hydraulic system shall not be performed unless the system is depressurized and has been emptied completely.
- Filters may not be changed unless there is ensured that the pump motor cannot be started.
- The settings of pressure relief valves effected by manufacturer shall not be changed, in order to prevent accidents.
- Quality, cleanliness and operating viscosity of the hydraulic medium are decisive for safety of operation, economy and service life of the machine.
- Use only the grade of hydraulic oil recommended by LOESCHE. The class of purity of the employed oil at least should conform to **NAS 1638 Class 8 or better.**

Hydraulic Cabinet HSMS ...

Structure and Function

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1. Introduction

The rollers of the LOESCHE-Mill are lubricated by circulating oil. The oil lubrication system is the hydraulic cabinet HSMS, whose position and dimension of the pipe connections are to be taken from the dimension sheet.

2. Installation

The lubricating cabinet should be located on a base and anchored with 4 screws. Make sure the rear of the cabinet is accessible as per the dimension sheet.

The lubricating pipes branch off at the rear of the hydraulic cabinet and are led through a foundation trench to the mill. Each connection is marked by an indicating letter. The same letter can be found on the Design Drawing "Lubrication Line, cpl." of the LOESCHE Mill.

The cooling-water connections are also on the rear of the hydraulic cabinet.

3. Design and Operation

The lubricating cabinet consists of a steel casing with a reservoir at the top with a total capacity of 850 l. On the front cover there are two oil sight glasses, one above the other.

During mill operation the oil level is near the middle of the lower sight glass. Do not fill any higher, since the oil inside the rollers flows back into the reservoir when the roller swings out. On top of the reservoir there is the filler/air vent. The oil filling can be emptied by means of the drain cock. A heater cartridge, mounted inside a finned tube, serves the purpose of heating up the contents of the reservoir. The heating capacity and the surface of the finned tube are designed so as to minimize the specific heat load per cm² of heating surface, in order to prevent "cracking" of the oil. Next to the finned tube a copper-clad steel capillary has been screwed in which is connected to the thermostat in the front panel of the cabinet. This allows pre-setting the required oil temperature.

Below the reservoir there are gear pumps arranged on each of the two levels which are situated one above the other. Each roller of the mill has its own supply pump and suction pump. Each of the lines connecting the reservoir with the pump can be closed by a shut-off valve.

The feeder pump has a capacity of 6 l/min; it sucks oil from the reservoir and pumps it through a filter and a line into the roller. The filter has an integral dirt trap to prevent damage to the filter insert. To protect the pump there is a relief valve set to 1 MPa (10 bar¹⁾) max. which opens if the oil in the lines is too viscous to be pumped due to low ambient temperature. The pressure can be read on a pressure gauge. Inside the lubrication cabinet HSMS the feed line to the roller is equipped with a water/oil heat exchanger. This prevents oil at too high a temperature from being pumped to the roller. To maintain a constant temperature on the oil the cooling water regulation valve is automatically controlled.

The oil level within the roller is determined by a radial bore in the roller axle. As soon as the oil level completely covers this opening of the hole, the suction pump builds up a vacuum in the suction line. With a capacity of 10 l/min, the suction pump causes the warm oil level inside the roller to drop faster than the supply pump can raise it. The suction pressure can be read on a vacuum gauge.

The oil returning from the roller can be observed in the sight glass. As soon as the oil level has dropped sufficiently to expose the hole in the roller axis, the vacuum in the return line collapses and the oil flow through the sight glass stops. The entire sequence is repeated as soon as the supply pump has raised the oil level above the suction line opening.

Although the pressure side of the suction pump is filled with oil from the reservoir by gravity, an "oil bleed" line with throttle is provided as an additional safeguard for adequate lubrication of this pump. This can be closed if the suction pump does not heat up appreciably.

4. Initial Start-Up

Refer to the drawings to make sure that the pipelines between the lubricating cabinet and the rollers have been correctly installed. All pipe fittings and connections must be carefully tightened, especially the oil return line. The vacuum required for oil suction can only be obtained in this pipe if no outside air can enter.

To fill the reservoir, when the rollers are empty, unscrew the cap and fill as high as the upper oil level. When the rollers are filled with oil the reservoir should not receive more than 400 l.

According to the bearing manufacturer's recommendations the lubricating oil should have a viscosity of 220 mm²/s at 40°C. In countries where the outside temperature does not drop below 20°C, such as the Central European area in the summertime, we recommend an oil 320 mm²/s at 40°C.

1) 1 bar = 14.5 psi

If the cabinet does not stand in the immediate vicinity of the mill, we recommend an electrical companion heating of the oil pipes in the foundation trenches in order to avoid an increase in oil viscosity due to the cooling of the pipelines as this could impair the proper functioning of the suction pump.

Dependent upon the distance between pump and roller and the viscosity of the oil in the tank and roller (in turn temperature dependent) the flow resistance in the lubrication system varies. The flow-limiting valve in the feed line to the roller is, on commissioning, adjusted to a low pressure (ca. 100 kPa to 200 kPa = 1 to 2 bar) to avoid overfilling the roller. By this means, the bulk of the viscous oil flows in a "short circuit" back into the tank. This circulation heats the oil. As the viscosity decreases, the flow limiting-valve shuts and more and more warm oil is fed to the roller. Here it is further warmed and thus enables the suction pump to function freely.

The temperature which has been set on the temperature monitor should not be exceeded. The temperature zone on the cooling water control valve will be chosen by the LOESCHE (commissioning) engineer during commissioning.

Normal working condition is achieved when the oil pulses regularly through the return line sight glass. It is then only necessary to adjust the flow-limiting valve when extreme ambient temperature differences exist between summer and winter.

5. Normal Operation

The mill motor, reservoir heater and lubricating pump motors are interlocked in such a way that the pumps cannot start until a required oil temperature of 50 - 60°C, preset at the temperature monitor, has been attained, and the mill motor has been running for about 20 to 30 minutes. In this way the reservoir oil is made ready for the pump, and the roller sump oil is sufficiently heated up for the suction pump to return it to the tank. Proper operation is indicated through the sight glass - as already described - when the return oil stream is not continuous but flows in a pulsating fashion.

6. Maintenance

Each supply pump is safeguarded by a relief valve set to appr. 100 to 200 kPa (1 to 2 bar).

During the first 8 hours following initial start-up, each filter insert should be checked hourly and replaced, if found to be contaminated. After that we recommend a weekly filter check and an exchange as necessary.

At the same time the air filters at the far ends of the roller shafts should be inspected. If cleaning is required use a petrol solution for washing them out and dip them in machine oil before reassembly.

After 3500 hours of operation or at the latest after one year, empty all oil from rollers and cabinet. To remove oil sludge from the bottom of the tank, unscrew the rear inspection cover.

In case the oil does not circulate properly, check the suction lines and the return line sight glasses for tightness as well as the oil temperature and viscosity. If the cool oil is too viscous, the only remedy is to heat it sufficiently and to reduce the pressure setting of the flow-limiting valve.

To check the suction line for tightness, proceed as follows:

Unscrew both hose connections at the end of the roller shafts. Let the free hose ends hang into a clean container filled with four or more litres of oil. Switch on the pumps. The oil level inside the container must be held at the level of the suction hose inlet connection. When the suction hose end is lowered, the oil level should drop again.

7. Swinging-Out Process

The lubrication lines permanently installed on the mill lower housing are connected by hoses with the oil distributor at the end of the roller shafts. This arrangement allows the oil supply to remain connected when the rollers swing out. At the same time the oil inside the rollers returns to the tank through the open throttle of the "oil bleed" line of the suction pumps.